

Area of Trapezoids

Lesson 5-2

Name: _____

Date: _____

Class: _____

Key Vocabulary

Level 1 support

Picture first, then the word, then a plain-language meaning. Say each word out loud.

A table top wider than the bottom shelf — the top and bottom edges are parallel, the two sides slant inward

Trapezoid

A four-sided shape with just one pair of parallel sides.

If the bottom of the trapezoid is 10 ft, then $b_1 = 10$ ft in the formula $A = \frac{1}{2}(b_1 + b_2) \times h$

Base 1 (b_1)

One of the two parallel sides of a trapezoid.

If the top of the trapezoid is 6 ft, then $b_2 = 6$ ft; both bases are parallel to each other

Base 2 (b_2)

The other parallel side of a trapezoid.

A dashed vertical line from the top base straight down to the bottom base at a 90° angle — NOT the slanted side

Height

The straight-up distance between the two parallel sides.

A trapezoid with $b_1 = 10$, $b_2 = 6$, $h = 4$ has area = $\frac{1}{2}(10 + 6) \times 4 = 32$ sq units

Area

How much space is inside a flat shape.

Key Ideas & Notes

- Your architecture firm is designing a trapezoidal window for a modern building.
- The glass supplier charges by the square foot, so the team must calculate the exact area.
- The window has a top edge of 4 feet, a bottom edge of 8 feet, and a height of 5 feet.
- Plot the vertices of a trapezoid with bases of 8 and 4, and height of 5. Place points at $(0, 0)$, $(8, 0)$, $(6, 5)$, and $(2, 5)$.

Think About It

- What shape is the window?
- How many parallel sides does it have?
- What measurements are given?

My Notes

Turn & Talk — Discussion Points

Level 1 support

Level 2

Talk with a partner about each prompt. If you need help getting started, use the Level 1 sentence stems. Ready for more? Try the Level 2 question.

LAUNCH

The trapezoidal window has a top edge of 4 feet and a bottom edge of 8 feet. Why does the area formula use BOTH bases instead of just one?

Level 1 support

Start here: We use both bases because the top and bottom are different lengths.

We use both bases because the top and bottom are ____.

Usamos las dos bases porque la parte de arriba y la de abajo son ____.

If I used only one base, the area would be ____.

Si usara solo una base, el área sería ____.

Word bank: trapezoid base 1 (b1) base 2 (b2) height area

Listen for: Students explain a trapezoid has two parallel sides of different lengths, so using only one would over- or under-count the area; strong answers mention averaging the bases.

Level 2

The wonder prompt asks what happens if both bases are equal. Predict the shape and explain how the trapezoid formula would simplify in that case.

- If b1 equals b2, the shape becomes a ____ because ____.
- The formula simplifies to ____, since ____.

EXPLORE

On the grid you copied the trapezoid (bases 8 and 4, height 5), flipped it, and joined it to the original. What shape formed, and how does that explain the $\frac{1}{2}$ in the formula?

Level 1 support

Start here: Two trapezoids make a parallelogram, so one trapezoid is half of it.

When I flip and join the trapezoid, I make a ____.

Cuando volteo y uno el trapecio, formo un ____.

I added the two bases first because ____.

Primero sumé las dos bases porque ____.

The area is $\frac{1}{2}$ times (8 plus 4) times 5, which equals ____.

El área es $\frac{1}{2}$ por (8 más 4) por 5, que es ____.

Word bank: **trapezoid** **parallelogram** **base 1 (b1)** **base 2 (b2)** **height**

Listen for: A strong answer names a parallelogram, links the $\frac{1}{2}$ to taking half of two combined trapezoids, and computes $\text{area} = \frac{1}{2} \times 12 \times 5 = 30 \text{ sq ft}$.

Level 2

Compare finding a trapezoid's area to averaging the two bases and multiplying by the height. Are these the same method? Justify with the window numbers.

- Averaging the bases gives ____, and multiplying by the height gives ____, which ____ the formula answer.
- These methods are ____ because ____.

CONNECT

The builder pours concrete for a trapezoidal driveway with edges of 20 ft and 12 ft and a height of 15 ft; one bag covers 6 sq ft. Walk through finding the number of bags.

Level 1 support

Start here: Find the trapezoid area first, then divide by what one bag covers.

The driveway area is $1/2$ times (20 plus 12) times 15, which equals ____ square feet.

El área de la entrada es $1/2$ por (20 más 12) por 15, que es ____ pies cuadrados.

He needs ____ bags because ____ divided by 6 equals ____.

Necesita ____ bolsas porque ____ entre 6 es ____.

Word bank: trapezoid base 1 (b1) base 2 (b2) height area

Listen for: Students compute $\text{area} = 240 \text{ sq ft}$, then $240 / 6 = 40$ bags, and explain why dividing by coverage gives the bag count.

Level 2

If the builder doubled only the height of the driveway, would the number of bags double? Justify using the formula.

- Doubling the height ____ the area because ____.
- So the number of bags would ____, since ____.

Guided Examples

Example 1

What is the area of a trapezoid with bases 6 cm and 10 cm, and height 4 cm?

Solution: $A = \frac{1}{2} \times (b_1 + b_2) \times h = \frac{1}{2} \times (6 + 10) \times 4 = \frac{1}{2} \times 16 \times 4 = 32$ sq cm.

Answer: A. 32 sq cm

Example 2

A trapezoid has an area of 45 sq ft, bases of 7 ft and 11 ft. What is the height?

Solution: $A = \frac{1}{2} \times (b_1 + b_2) \times h \rightarrow 45 = \frac{1}{2} \times 18 \times h \rightarrow 45 = 9h \rightarrow h = 5$ ft.

Answer: A. 5 ft

Example 3

What is the first step when finding the area of a trapezoid?

Solution: The formula is $A = \frac{1}{2} \times (b_1 + b_2) \times h$. The first step is to add the two parallel bases.

Answer: A. Add the two bases together

Write About the Math

The Writing Revolution

I can explain my steps using the words trapezoid, base, height, and area.

1. Kernel Sentence subject + verb

Model: Trapezoid is a four-sided shape with just one pair of parallel sides.

Trapecio es una figura de cuatro lados con solo un par de lados paralelos.

Write a kernel sentence about trapezoid. Use a subject and a verb.

Escribe una oración base sobre trapecio. Usa un sujeto y un verbo.

2. Sentence Expansion because • but • so

Kernel: Trapezoid matters in math

Trapecio importa en matemáticas

Expand the kernel three ways. Add a reason, a contrast, and a result.

because
porque

Trapezoid matters in math because ____.

Trapecio importa en matemáticas porque ____.

but
pero

Trapezoid matters in math, but ____.

Trapecio importa en matemáticas, pero ____.

so
entonces

Trapezoid matters in math, so ____.

Trapecio importa en matemáticas, entonces ____.

3. Sentence Types 4 ways to write a math idea

Statement *Afirmación*

Tell one true fact about trapezoid.
Di un hecho verdadero sobre trapezoid.

Trapezoid ____.

Question *Pregunta*

Ask a question about trapezoid.
Haz una pregunta sobre trapezoid.

How does ____ ?

¿Cómo ____ ?

Exclamation *Exclamación*

Show excitement about trapezoid.
Muestra entusiasmo sobre trapezoid.

Wow, ____ !

¡Guau, ____ !

Command *Mandato*

Tell a partner what to do with trapezoid.
Dile a un compañero qué hacer con trapezoid.

First, ____ .

Primero, ____ .

4. Explain Your Reasoning use a sentence starter

We use both bases because the top and bottom are ____.

Usamos las dos bases porque la parte de arriba y la de abajo son ____.

If I used only one base, the area would be ____.

Si usara solo una base, el área sería ____.

When I flip and join the trapezoid, I make a ____.

Cuando volteo y uno el trapecio, formo un ____.

Try It

Solve on your own. Check the answer key when you are done.

1. What is the first step when finding the area of a trapezoid?

- A. Add the two bases together
- B. Multiply the two bases
- C. Subtract the shorter base from the longer base
- D. Divide the height by 2

Show your work:

2. A trapezoid has bases of 5 in and 9 in, and a height of 6 in. What is its area?

- A. 42 sq in
- B. 84 sq in
- C. 27 sq in
- D. 30 sq in

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

A trapezoid and a parallelogram both have a height of 8 ft. The trapezoid has bases of 5 ft and 11 ft. The parallelogram has a base of 8 ft. Which shape has the greater area? Explain your reasoning using the formulas.

Sentence starter: The trapezoid's area is ____ because $\frac{1}{2} \times (\text{ } + \text{ }) \times \text{ } = \text{ }$. The parallelogram's area is ____ because $\text{ } \times \text{ } = \text{ }$. The ____ has a greater area by ____ sq ft.

Show your work:

Reflect — Exit Ticket

A trapezoid has bases of 9 inches and 5 inches, and a height of 6 inches. What is its area?

- A. 42 sq in
- B. 84 sq in
- C. 27 sq in
- D. 42 in

Your answer:

Answer Key & Teacher Guide

1. **Try It 1:** A. Add the two bases together — *The formula is $A = \frac{1}{2} \times (b_1 + b_2) \times h$. The first step is to add the two parallel bases.*
2. **Try It 2:** A. 42 sq in — $A = \frac{1}{2} \times (5 + 9) \times 6 = \frac{1}{2} \times 14 \times 6 = \frac{1}{2} \times 84 = 42 \text{ sq in.}$
3. **Exit Ticket:** A. 42 sq in — $A = \frac{1}{2} \times (b_1 + b_2) \times h = \frac{1}{2} \times (9 + 5) \times 6 = \frac{1}{2} \times 14 \times 6 = 42 \text{ square inches.}$

Writing (TWR) — what to look for

- **Kernel sentence:** A complete sentence needs a subject and a verb. Example: Trapezoid is a four-sided shape with just one pair of parallel sides.
- **Expansion:** *because* gives a reason, *but* shows a contrast or exception, *so* shows a result. Answers vary; each must keep the kernel idea and add the correct kind of detail.
- **Sentence types:** Statement ends with a period, question with "?", exclamation with "!", and a command starts with an action verb (a "bossy" verb).